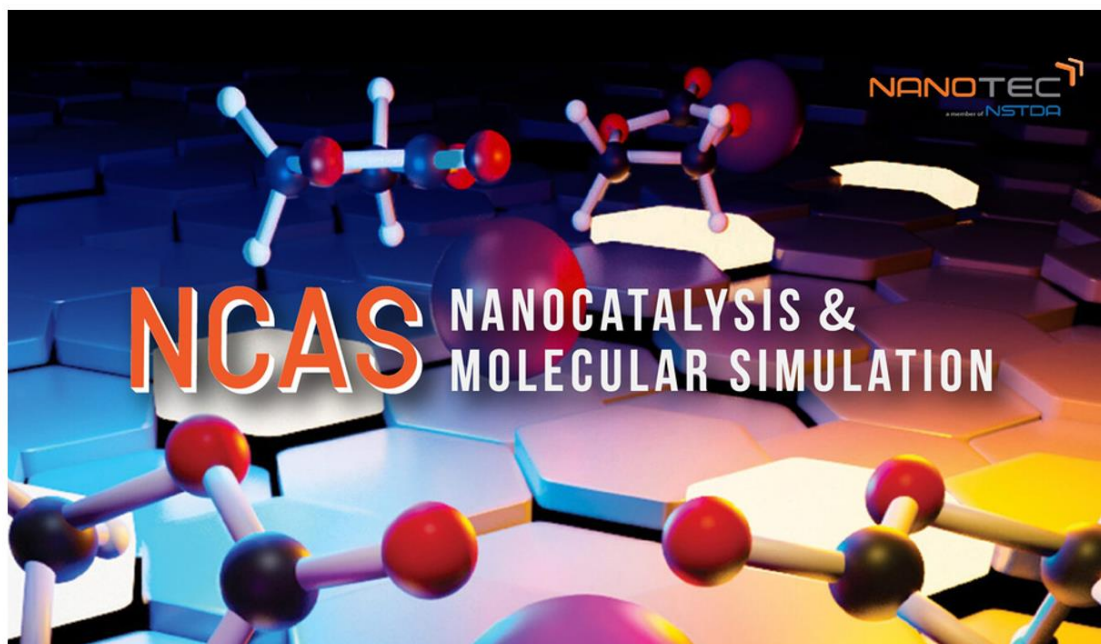
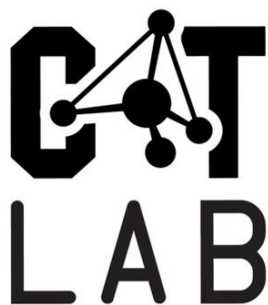




Project Title:

Development of Cu-based catalyst for electrochemical CO₂ reduction to C₂ products: combined computational chemistry and electrochemical engineering approach



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1. Research motivation

Carbon dioxide (CO_2) is a major greenhouse gas that contributes to global warming and climate change. Reducing CO_2 emissions from fossil fuel combustion is an urgent challenge for environmental protection and energy sustainability. One promising strategy is to directly capture CO_2 from flue gas and convert it into valuable chemicals (CO , HCOOH , CH_4 , etc.) electrochemical CO_2 reduction reaction (eCO_2RR). The C_1 hydrocarbon product plays a crucial role for the synthesis of high value-added chemicals and clean fuels.¹ Hence, significant endeavors have been committed to advancing the efficiency of electrocatalysts, aiming to decrease overpotential, enhance Faradaic efficiency (FE) for carbon monoxide (CO), and increase current density. This concerted effort has resulted in notable recent advancements.² In previous attempts to advance the CO_2 reduction reaction (CO_2RR), electrocatalysts have traditionally demanded a CO_2 feedstock of high purity to achieve optimal performance.³ However, the conventional CO_2 capture process involves multiple steps with a high energy cost.⁴ To implement CO_2RR on an industrial scale, it is preferable for the CO_2 feedstock to be derived from abundant sources like flue gas or air, simplifying the Carbon Capture and Utilization (CCU) process without the need for extensive energy in CO_2 purification.⁵ Nevertheless, the CO_2 concentration in flue gas is typically less than 15%, often accompanied by a certain amount of oxygen and other minor impurities.⁶ Consequently, electroreduction of flue gas encounters challenges such as limited CO_2 mass transfer and competition with the oxygen reduction reaction (ORR) due to the thermodynamic favorability of ORR.⁷ Therefore, there is a need to develop efficient electrocatalysts capable of both capturing CO_2 and inhibiting ORR for effective reduction of flue gas. Basically, not all porous catalysts can efficiently capture and enrich CO_2 under simulated flue gas conditions (15% CO_2 :85% N_2 , humidity of 40-80%) due to their weak CO_2 affinity as compared to H_2O molecules. To address this, strong capture CO_2 sites to serve as the “ CO_2 relays” are needed for porous catalysts to simultaneously capture and enrich CO_2 molecules in the pores of a catalyst (as shown in Fig. 1.1), which then travel to the catalytic active sites for eCO_2RR . Therefore, resulting in a high catalytic performance under diluted CO_2 gas stream (flue gas).⁸

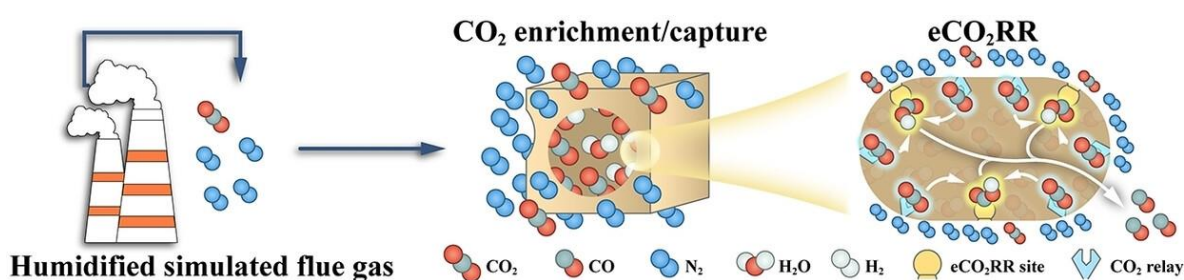


Figure 1.1 Simultaneous CO_2 capture and eCO_2RR in amine-functionalized Ag_{12} cluster-based MOF ($\text{Ag}_{12}\text{bpy-NH}_2$) cathode catalyst boosts the catalytic performance under humidified simulated flue gas (15% CO_2 :85% N_2).⁸

Presently, Liu Y et al.⁸ reported the simultaneous capture and utilization of CO_2 by integrating CO_2 capture capability and eCO_2RR catalytic active sites in the micropores of Ag_{12} cluster-based metal-organic framework ($\text{Ag}_{12}\text{bpy-NH}_2$). Under a simulated flue gas, the $\text{Ag}_{12}\text{bpy-NH}_2$ catalyst exhibited commercial catalytic performance, which was close to its performance in a pure CO_2 atmosphere and even higher than other catalysts. These results indicate that the less affected eCO_2RR performance of $\text{Ag}_{12}\text{bpy-NH}_2$ under a diluted CO_2 gas stream (flue gas) might be attributed to the addition of CO_2 relay (amine groups) around Ag atoms that could potentially enrich CO_2 and subsequently migrate to active sites of Ag_{12} .

clusters, where CO₂ molecules get chemically reduced to CO. on the other hand, Liu Z et al.⁹ investigated the simultaneous CO₂ capture and utilization of CO₂ to CO under simulated flue gas (15%CO₂:4%O₂:81%N₂) with pristine Ag nanoparticles and amine-functionalized ligand layer on Ag nanoparticles. The results indicated that the amine-functionalized Ag NPs exhibited a FE_{CO} of 82%, while a pristine Ag NPs achieved FE_{CO} of 31.8% under the same condition in H-cell. Besides the H-cell, the amine-functionalized Ag NPs also achieved high FE_{CO} of 79.6% at 2.8V and 48 mA/cm² in a zero-gas MEA electrolysis cell. These findings demonstrate that the amino functional groups play a crucial role in facilitating the CO₂ capture and activation, which then migrate to the catalytic active sites of Ag to chemical convert CO₂ into CO product.

Inspired by these above concepts, the functionalization of Ag catalyst surface with high CO₂ capture capability such as amine (-NH₂), PANI (polyaniline), or other functional groups, could potentially enhance the electrocatalytic activity, selectivity, and MEA electrolyzer efficiency under a low CO₂ concentration gas stream.

2. Objective

- Synthesis of functionalized Ag-based cathode catalyst for simultaneous CO₂ capture and utilization processes.
- Studies of electrochemical performance and stability of AEM electrolysis cell towards eCO₂RR into CO product under various operating conditions.

3. Scope of research

- Synthesis of bare-Ag and functionalize-Ag catalysts and hydrothermal methods.
- Characterization of cathode samples by XRD, SEM, TEM, and XPS to physically and chemically determine optimal morphology and Ag nanoparticle size.
- Fabrication of Ag-gas diffusion electrodes using sputtering or spray coating technique with variables such as the catalyst loading, and amount of binder mixed in the ink.
- MEA cells construction with NiFe foam anode, sustainion anion exchange membrane, and as-prepared Ag-based cathode catalyst, respectively.
- Electrochemical measurements of CO₂ reduction cathode catalysts in MEA cells under humidified CO₂, varying current density (50-200 mA/cm²), type and concentration of anolyte to determine the optimal condition for maximizing CO production efficiency.
- Electrochemical performance test of CO₂ reduction cathode catalyst under simulated flue gas stream such as N₂, O₂, SO_x, and NO_x gas impurities to investigate the influence of gas impurities concentrations on selectivity and efficiency of the product.

Main tasks for November:

1. To have a grasp on AEM electrolysis cell system by using commercial Ag cathode, IrO₂ anode, and Sustainaion membrane.
2. To study AEM electrolysis cell reproducibility towards eCO₂RR to CO under operating conditions.

3. Experimental procedure

3.1 AEM cell installation

A complete 5 cm² CO₂ electrolyzer as shown in Figure 1(a) consists of: Titanium anode flow field, 904 L Stainless Steel cathode flow field, catalyst covered electrodes for anodes and cathodes of 5 cm² cell, Sustainion® membrane, nuts, bolts, o-rings and insulating kit. Figure 1(b) demonstrates the CO₂ electrolyzer operating system configuration.

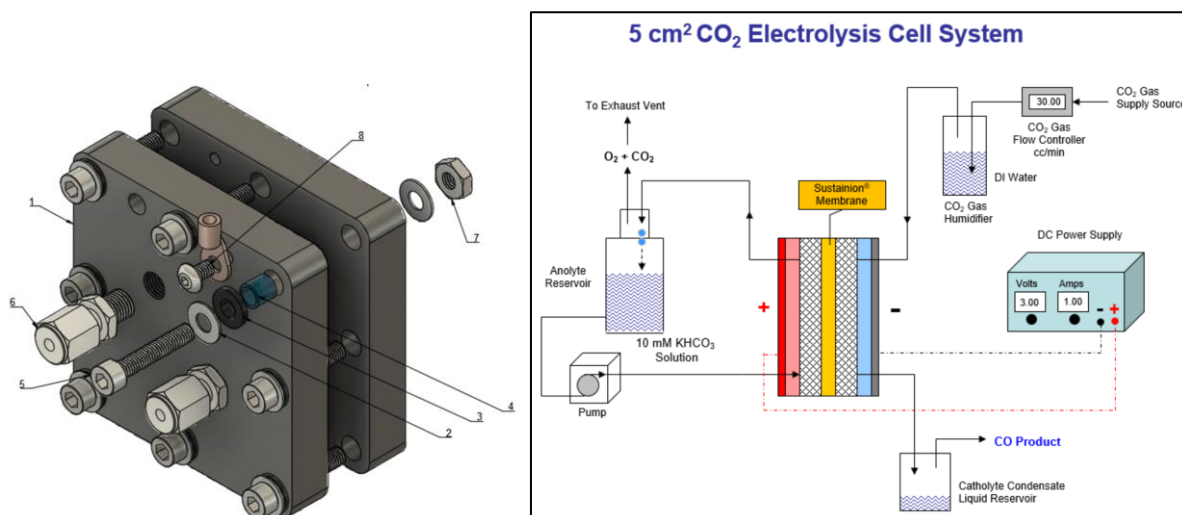


Figure 1. (a) DM 5 cm² AEM CO₂ electrolyzer configuration and (b) electrolyzer setup.

AEM cell installation procedure was carefully followed through company's instructions:

- Position the Anode flow field plate (Titanium anode) upright with flow field facing laterally (#1) and threaded holes (#8 and 9) on top. #8 and 9 is for electrical wire/cables and thermal couple. Attach stainless steel flat washer (#2), insulating shoulder washer (#3) and insulating tube (#4) in that sequence to 10-32 × 1" socket head cap screw (#5). Repeat this procedure for all 8 bolt holes.
- Lay flat (with all bolts pointing up and flow field exposed), place O-ring into groove for O-ring on the anode flow field and gasket for anode (with 3.1cm × 3.1cm square hole).
- Place the anode (3 × 3 cm) in the square hole (3.1 X 3.1 cm) of the anode gasket. Do not overlap the anode with the gasket.
- Prepare one piece of 5 × 5 cm membrane and place the membrane on top of the anode with the anode in the center of the membrane.
- Place the cathode gasket with eight round holes (for screws) and one square hole (for cathode) on top of the membrane, and locate cathode (2.5cm × 2.5cm) in the square hole (2.6 cm × 2.6 cm) of the gasket. Do not overlap the cathode with the gasket.
- Place O-ring into O-ring groove on the cathode flow field (you can add several drops of DI water in the O-ring groove to prevent O-ring falling) and then place the cathode flow field with flow field facing down on top of the assembly. Channels in cathode flow field are parallel to those in anode flow field.
- Insert flat washers and nuts (#6/7) and tighten the cell diagonally up to 60 in-lb using torque wrench and Allen key.

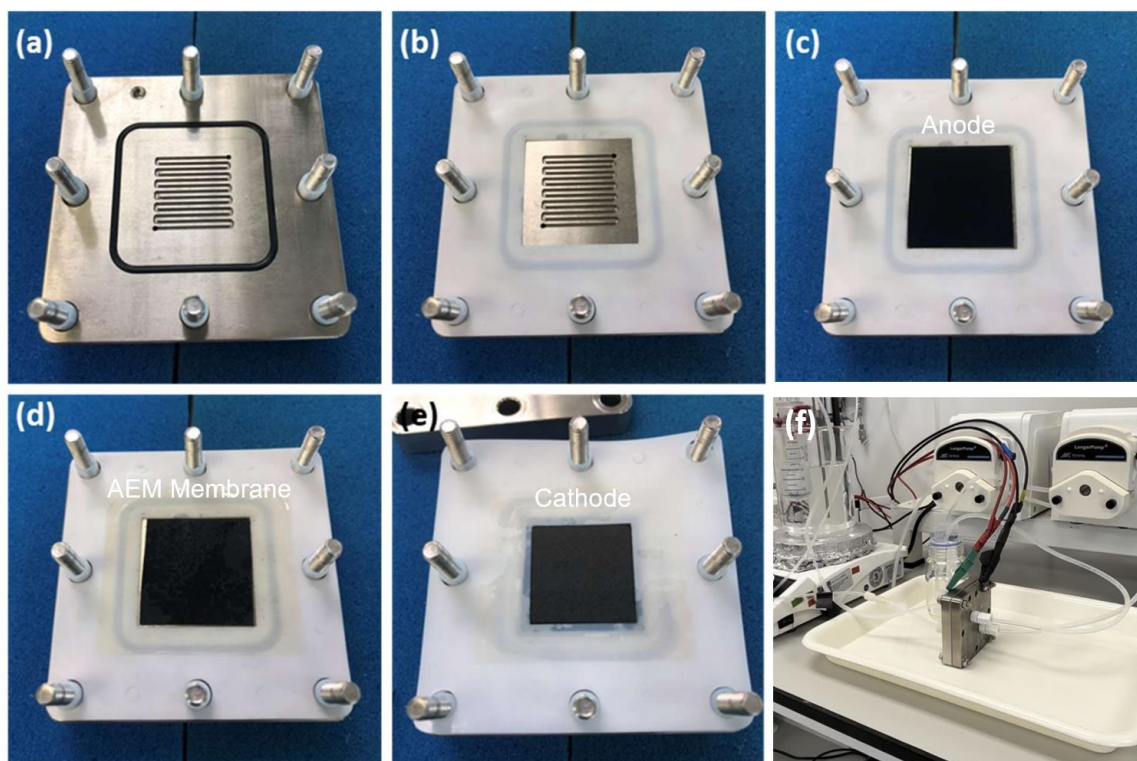


Figure 2. AEM electrolyzer installation process.

3.2 Electrochemical measurements and products analysis

The electrochemical measurements such as electrochemical impedance spectroscopy (EIS) and chronopotentiometry (CP) of the AEM electrolyzer were conducted using electrochemical workstation (Bio-logic, HCP-803) under 30 ml/min of humidified CO₂ gas stream, 3 ml/min of 10 mM KHCO₃ anolyte, varying $T_{\text{cathode-stream}}$ (25, 40, 60°C), and applied current density (50, 100, 150, 200 mA/cm²). In CP mode, the cell voltage was continuously recorded, and the cathodic gas products (Mainly CO and H₂) were periodically analyzed under a specific applied current density by Gas chromatography (GC), while the volumetric flow rate of cathodic outlet stream was also measure by a flow meter that will be further use for %FE calculation. After 1 hour of the operational testing, the cathode compartment was flushed out by flowing DI water (for up to 5 min) to prevent salt formation over time.

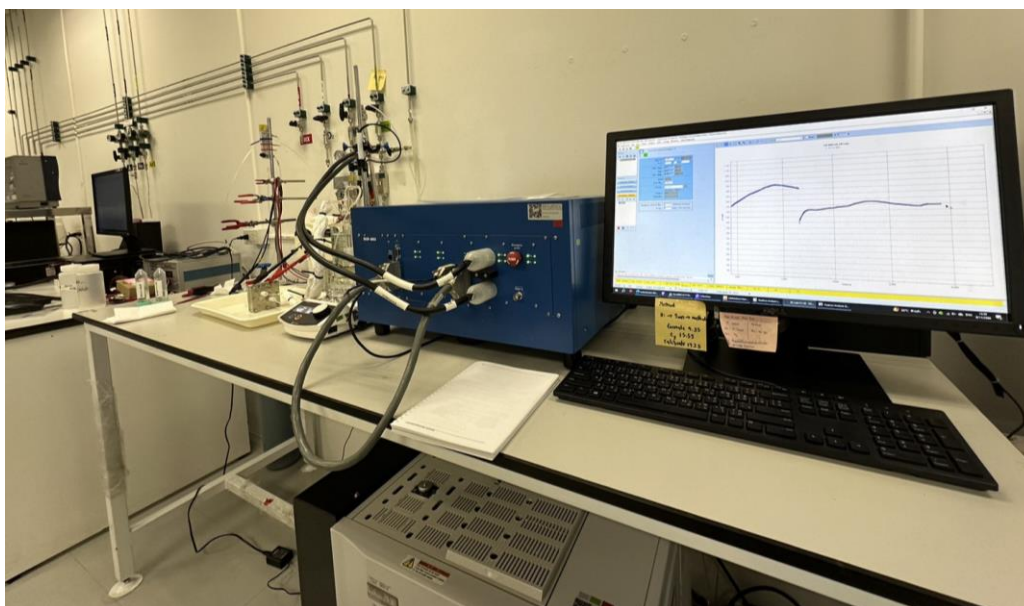


Figure 3. Electrochemical workstation (Bio-logic, HCP 803).

4. Results

Influence of current density on faradaic efficiency

In the MEA electrolysis cell, current density (j) plays a crucial role in the production rate and selectivity of the products. As shown in Fig. 5.1, we observed a gradual increase in %FE(CO) at high current densities, while FE(H₂) remained steady under 50-100 mA/cm² and marginally increased to ~2.5% under 150-200 mA/cm². These results indicate the dominance of eCO₂R to CO in the current density range of 50-200 mA/cm². However, the total faradaic efficiency of the products (a combination of %FE(CO) and %FE(H₂)) was much lower than 100%. This might be attributed to CO₃²⁻, partial HCO₃⁻, and HCOO⁻ diffusing across the AEM membrane to the anolyte, thereby reducing CO₂ reactant at the cathode side. Additionally, CO₂ crossover leads to a nonstoichiometric decrease in the outlet flow rate from the cathodic compartment. This effect can result in a substantial overestimation of catalytic performance if the outlet flow rate of CO₂ is used as a reference for calculating Faradaic efficiencies.¹⁰

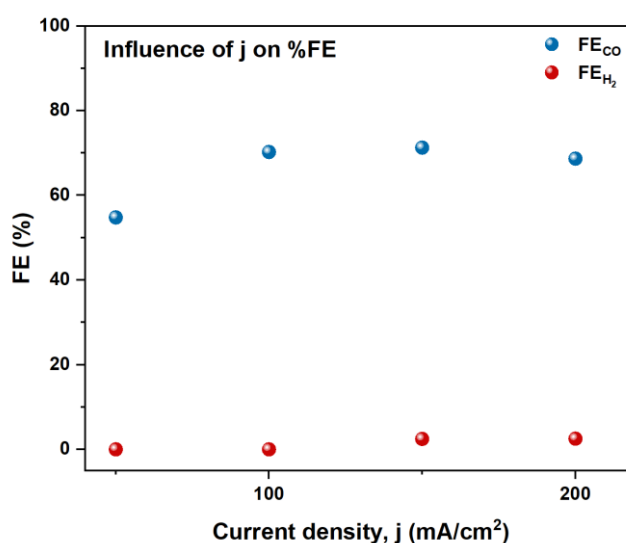


Figure 5.1 Faradaic efficiency of CO and H₂ under varying current density of 50-200 mA/cm²

Influence of CO₂ gas stream temperature on faradaic efficiency

Water availability in the catalyst layer is necessary for facilitating the CO₂ reduction reaction (CO₂RR). However, an excess of water promotes the competing hydrogen evolution reaction (HER) and diminishes the effectiveness of CO₂ electrolysis. As shown in Fig. 5.2, %FE(CO) was slightly increased from 74% to 65% after raising CO₂ gas stream temperature from 30 to 60 °C. This might be attributed to an increase in humidity at the cathodic CO₂ stream in which excess water can cause HER to be preference over eCO₂R.¹¹ Furthermore, the results clearly indicated an increase in %FE(H₂) at high CO₂ gas stream temperature caused by HER preferential.

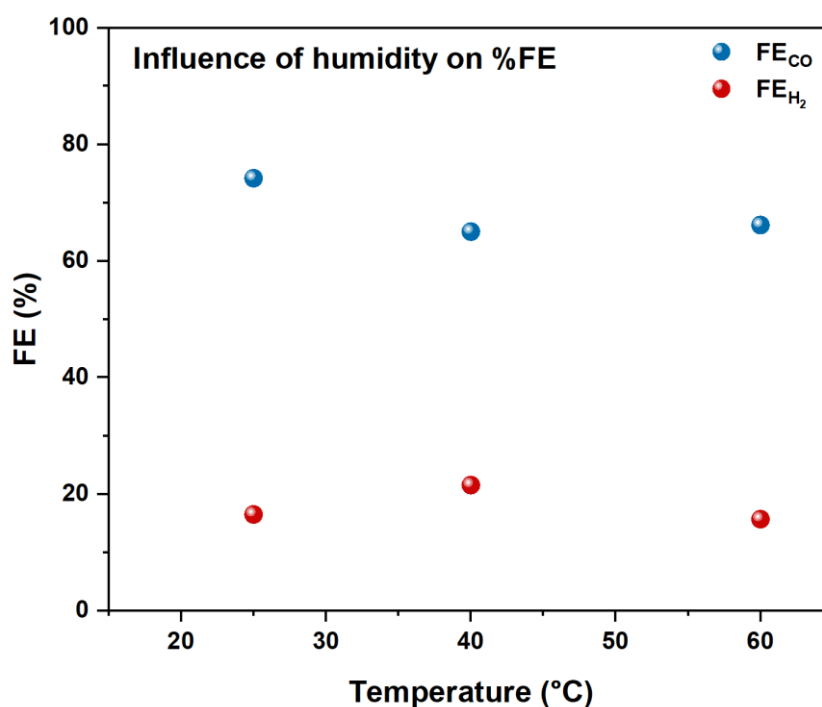
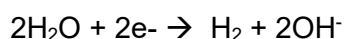
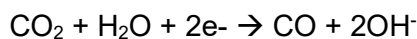


Figure 5.2 Faradaic efficiency of CO and H₂ under varying T_{cathodic stream} of 20-60 °C.

Faradaic efficiency of AEM cell under 3 consecutive operational test days

To assess the stability of the AEM cell in terms of the Faradaic efficiency (FE) for both CO and H₂, we conducted consecutive tests spanning over 3 days, comparing the %FE for each day. As the results indicate, %FE(CO) significantly decreased over time from ~97% to 57%, while %FE(H₂) gradually increased from 14% to 25%. Moreover, the current density substantially decreased over time at 3.5 V, and these findings suggest a decline in the eCO₂R performance of the AEM cell over the course of the extended test period.

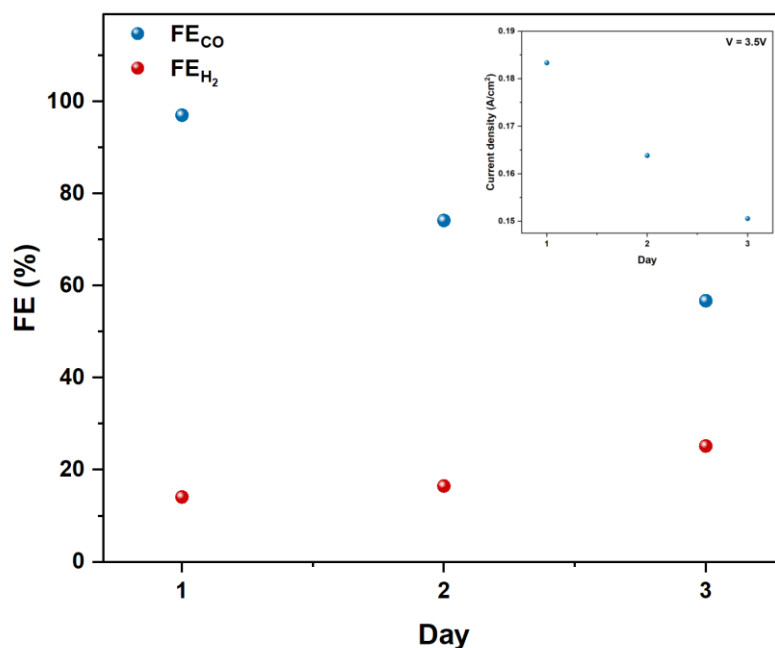


Figure 5.3 Faradaic efficiency of CO and H₂ under applied voltage of 3.5V and 3 consecutive operational test days.

5. Summary

In summary, the AEM cell was successfully assembled with components of Ag-GDE cathode, Sustainaion membrane, and IrO₂ anode. We found that high current densities favored eCO₂R to CO products with a fraction of H₂, and FE(total) was below 100% which attributed to membrane diffusion, CO₂ crossover effects, and some other issues that may need to be addressed. Additionally, excess water preferred competing reactions (HER), leading to increase %FE(H₂) at high CO₂ gas stream temperature. The operational stability over 3 consecutive days demonstrated a significant decrease in %FE(CO), and an increase in %FE(H₂), indicating a decline in eCO₂R performance.

6. Upcoming tasks

- Literature review on the synthesis of Ag nanoparticles and functionalized Ag-based catalyst.
- Experimental design for the synthesis part.
- Reproduce the result under the same conditions with new NiFe foam anode and bipolar membrane.

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